

Catastrophic Forgetting and the Pseudorehearsal Solution in Hopfield-type Networks

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(Received for publication April 1997; revised paper accepted January 1998)

Pseudorehearsal is a mechanism proposed by Robins which alleviates catastrophic forgetting in multi-layer perceptron networks. In this paper, we extend the exploration of pseudorehearsal to a Hopfield-type net. The same general principles apply: old information can be rehearsed if it is available, and if it is not available, then generating and rehearsing approximations of old information that 'map' the behaviour of the network can also be effective at preserving the actual old information itself. The details of the pseudorehearsal mechanism, however, benefit from being adapted to the dynamics of Hopfield nets so as to exploit the extra attractors created in state space during learning. These attractors are usually described as 'spurious' or 'cross-talk', and regarded as undesirable, interfering with the retention of the trained population items. Our simulations have shown that, in another sense, such attractors can in fact be useful in preserving the learned population. In general terms, a solution to the catastrophic forgetting problem enables the on-going or sequential learning of information in artificial neural networks, and consequently also provides a framework for the modelling of lifelong learning/developmental effects in cognition.

KEYWORDS: Catastrophic forgetting, catastrophic interference, consolidation, rehearsal, pseudorehearsal, Hopfield network.

1. Introduction

Catastrophic forgetting is the tendency of newly learned information (a new input/output population of patterns) to disrupt or eliminate the memory of old information (any previously learned population) in a network. Grossberg (1987) suggests the analogy of a human trained to recognize the word *cat*, and subsequently to recognize the word *table*, being then unable to recognize *cat*. Catastrophic forgetting is a well-known and significant problem for artificial neural network (ANN) learning, making it very difficult to integrate new information into an existing network when using a broad range of common ANN learning algorithms such as, for example, backpropagation (Rumelhart *et al.*, 1986).

The catastrophic forgetting problem is the reason that most ANN learning

algorithms involve concurrent learning, i.e. the whole population of interest is presented and trained (over a number of iterations) as a single, complete entity. Training is then 'finished' and no further information is learned by the network. There is no natural and effective way to train sequentially, i.e. have multiple training sessions where some learning is completed before other learning begins. Concurrent learning is undesirable in practical terms, making it very difficult to modify or extend any given ANN application without completely retraining the network (compared with a traditional rule-based system where information or rules can easily be added to or removed from the system). It is also a highly implausible constraint for ANN-based cognitive models where so much of human learning is clearly sequential or incremental in nature. On-going or 'lifelong' learning/development and any kind of transfer effects that add new information to a system cannot be modelled at all without addressing the catastrophic forgetting problem in some way.

Pseudorehearsal is a solution to catastrophic forgetting described by Robins (1995) and developed in Robins (1996), French (1997) and Freen and Robins (1997). This method has so far been explored only in multi-layer perception (MLP) networks. Here, we extend the investigation of pseudorehearsal to Hopfield-type networks. Rehearsal and pseudorehearsal methods allow new information to be added to a network without disrupting old information, providing both a practical solution to catastrophic forgetting, and a foundation for modelling consolidation, on-going learning and a range of transfer effects in ANNs.

We begin by briefly reviewing in Section 2 the work on catastrophic forgetting, rehearsal and pseudorehearsal in MLP networks. In Section 3, we describe a Hopfield-type network that uses the thermal perceptron learning algorithm (Freen, 1992) to achieve a high storage capacity. In Sections 4 and 5, we explore the catastrophic forgetting problem, and the rehearsal and pseudorehearsal solutions, using the Hopfield-type network. Both rehearsal and pseudorehearsal are effective at reducing catastrophic forgetting, but the specifics of the pseudorehearsal mechanism developed for MLP networks benefit from being adapted to the dynamics of Hopfield nets.

2. Catastrophic Forgetting, Rehearsal and Pseudorehearsal in MLP Networks

2.1. Catastrophic Forgetting

One consequence of a failure to address the 'stability/plasticity dilemma' (Carpenter & Grossberg, 1988; Grossberg, 1987) in many neural networks is excessive plasticity, usually called 'catastrophic forgetting' (or 'catastrophic interference' or the 'serial learning problem'). If a network is exposed to the learning of new information, then any previously learned information will typically be greatly disrupted or lost.

While stability/plasticity issues are very general, the term 'catastrophic forgetting' has tended to be associated with 'static' networks (rather than 'dynamic' nets which add or remove nodes) employing supervised learning. A number of recent studies have used MLP-type networks (typically employing backpropagation) to highlight the problem of catastrophic forgetting and explore various issues (these include Freen & Robins, 1997; French, 1992, 1994, 1997; Hetherington & Seidenberg, 1989; Lewandowsky, 1991; Lewandowsky & Li, 1995; McCloskey &

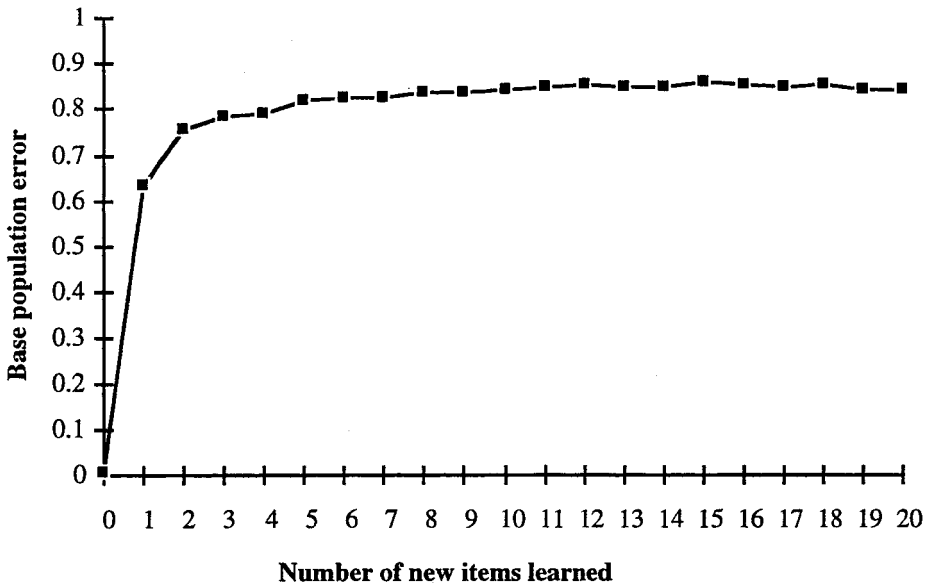


Figure 1. The basic catastrophic forgetting effect in an MLP network. The base population error measure is the average sum squared error between actual and target outputs (adapted from Robins (1996), Figure 1).

Cohen, 1989; McRae & Hetherington, 1993; Murre, 1992; Ratcliff, 1990; Robins, 1995, 1996; Sharkey & Sharkey, 1995). Similar issues have been explored in the context of Hopfield networks by Nadal *et al.* (1986) and Burgess *et al.* (1991).

In a typical illustration of catastrophic forgetting,¹ we use a backpropagation network to learn a base population of items (input/output vector pairs) in the usual way (Rumelhart *et al.*, 1986). Subsequently, a number of further items are learned one by one—these are the ‘new’ trials. The effect of these new item trials can be illustrated by plotting a measure (such as goodness or error) or the ability of the network to reproduce correctly the base population after each new item. As shown in Figure 1, the error in a base population of items increases ‘catastrophically’ as new items are learned.

2.2. Rehearsal

A range of solutions to this problem has been explored in the studies noted earlier, focusing in particular on either reducing the overlap of hidden-unit representations or rehearsing old items as new items are learned. Here, we focus on the rehearsal-based solutions as explored in Hetherington and Seidenberg (1989), Ratcliffe (1990), Murre (1992) and Robins (1995, 1996). A ‘broad-but-shallow’ rehearsal regime, rehearsing many old items as each new item is introduced but not necessarily training them to criterion, appears to be the most effective rehearsal strategy. Repeating the learning task described in Figure 1 using rehearsal as each new item is learned eliminates the catastrophic forgetting effect, as shown in Figure 2. (Specifically, Figure 2 illustrates the ‘sweep rehearsal’ regime described in Robins (1995, 1996).)

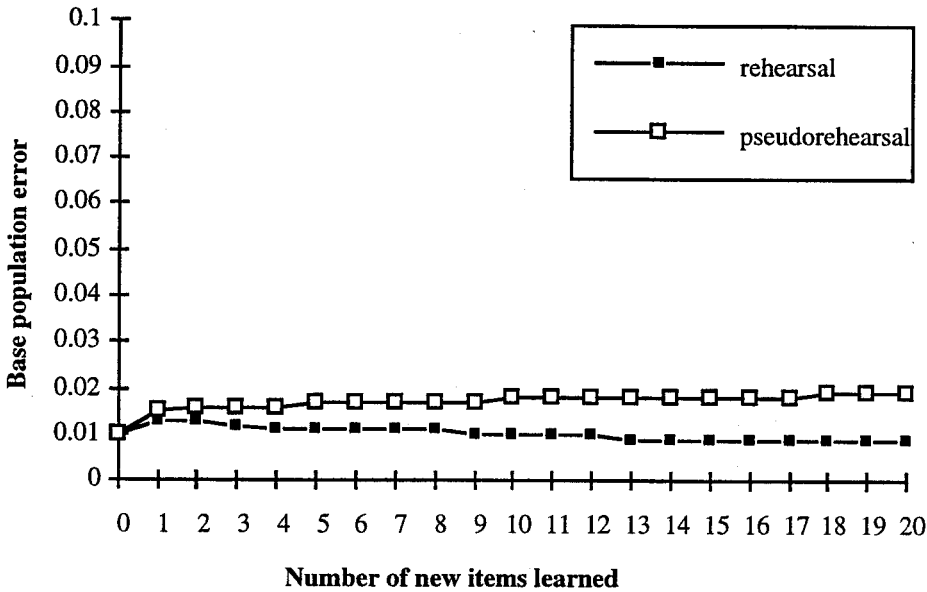


Figure 2. Rehearsal and pseudorehearsal methods in an MLP network. The base population error measure is the average sum squared error between actual and target outputs (adapted from Robins (1996), Figures 3 and 4; note the different scale on the y-axis cf. Figure 1).

2.3. Pseudorehearsal

As a practical methodology for dealing with catastrophic forgetting, simple rehearsal is very effective—as long as the previously learned items are actually available for relearning. It may be, however, that the old items have been lost, or it is not practical for some reason to store them. Sharkey and Sharkey (1995) note, for example, that:

... the interference [catastrophic forgetting] problem is [...] general and should be of concern to all those involved in developing applications in which the training data only become available piecemeal over an extended period of time. For example, in on-line learning of control processes, such as found in robotics or manufacturing, it may not be practical to maintain all of the training data in memory and retrain each time a novel aspect of the data is encountered (p. 302).

Similar arguments seem particularly relevant in the context of modelling cognition.

We can still achieve most of the benefits of rehearsal, however, even when we do not have the old items to rehearse (i.e. even when there is no access to the base population). Pseudorehearsal (Robins, 1995) is based on the use of artificially constructed populations of ‘pseudoitems’ instead of the actual previously learned items in the rehearsal process.

A pseudoitem is constructed by generating a new input vector at random, and passing it forward through the network in the standard way. Whatever output vector this input generates becomes the associated target output. For a given network (trained on a base population), a population of pseudoitems constructed in this way can be used instead of the actual base population items in any rehearsal regime. Repeating the learning task described in Figure 1, using pseudorehearsal

as each new item is learned, almost eliminates the catastrophic forgetting effect, as shown in Figure 2 (specifically, Figure 2 illustrates the 'sweep pseudorehearsal' regime described in Robins (1995, 1996)). Pseudorehearsal remains highly effective at preserving performance on the base population despite the fact that no actual base population items are rehearsed.

In short, rehearsing old items as new items are learned can protect previously learned information from catastrophic forgetting. Pseudorehearsal is a promising method for achieving most of the benefits of rehearsal in reducing catastrophic forgetting by approximating old information whenever it is needed (rather than assuming access to previously learned items for later rehearsal). As well as auto-associative learning with the Iris data set summarized here (from Robins, 1996) pseudorehearsal has so far been shown to be effective on randomly constructed auto-associative and hetero-associative data sets by Robins (1995), and in a classification task on the mushroom data set (Murphy & Aha, 1994) by French (1997).

In Robins (1995), we interpret the pseudorehearsal mechanism in terms of a function approximation-based explanation of the backpropagation network (see, for example, White (1992)). In learning the base population, the network fits some function to the base population 'data points' (population items). Rehearsal reuses these data points to preserve the shape of the function during the subsequent learning of new items. Pseudorehearsal samples the function at random places to construct new data points, which are then used in the same way to preserve the original function. In Frean and Robins (1997), we provide a formal analysis of a simple case of pseudorehearsal and describe situations in which pseudorehearsal can be proved to be effective.

Whereas these studies have been conducted in MLP networks, in the following we explore catastrophic forgetting, rehearsal and pseudorehearsal in a Hopfield-type network.

3. A Hopfield-type Network and Training Procedure

The network used in all simulations reported here consists of 64 units. Units have a threshold activation function and are switched on (activation of 1) if their net input exceeds 0, and off (activation of 0) otherwise. Units are randomly and asynchronously updated. For our purposes, we wish to study the storage of many patterns in the network; to achieve this, we use some generalizations of the standard Hopfield assumptions.

A standard Hopfield network using Hebbian learning is able to store at best roughly $0.138N$ arbitrary patterns (where N is the number of units in the network)—see, for example, the discussion in Hertz *et al.* (1991, pp. 17–20). The theoretical capacity of such a network, however, is $2N$ patterns (Cover, 1965; Gardner, 1987). The error-correcting delta rule can be used in Hopfield networks and increases the storage capacity to this theoretical limit of $2N$ (Gardner *et al.*, 1989), but fails to converge if the patterns cannot be stored in a network of the given size. A simple variant of the delta rule known as the 'thermal' perception learning rule (Frean, 1992) suffices to ensure convergence in all cases.² Consequently, we use the thermal perceptron rule in all simulations described in the following, with the learning constant α set initially to 1 and the temperature set initially to 128.

Note that the Hebbian learning rule leads to symmetric weights ($w_{ij} = w_{ji}$),

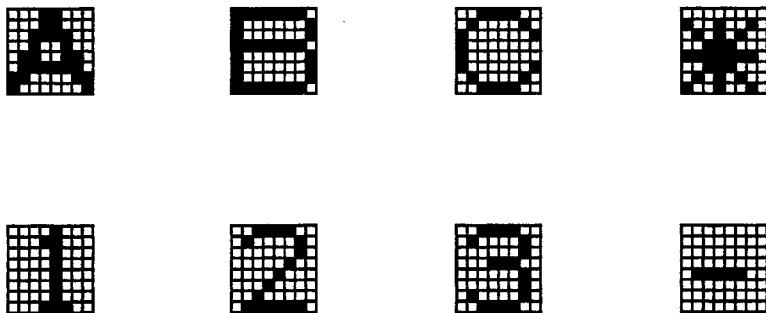


Figure 3. Eight examples from the set of 64 sample base population items (8×8 images of alpha-numeric characters).

which in turn ensure that stable attractor states and no cycles exist in the state space of the network. More powerful learning rules, such as the thermal perceptron algorithm, produce asymmetric weights. Hertz *et al.* (1991, pp. 48–49) briefly review the impact of asymmetric connections, noting that asymmetry effectively introduces some ‘noise’, and may slow down the approach to an attractor. If the asymmetry is ‘systematic’ or ‘very strong’ then it can produce limit cycles or chaotic behaviour, but this was not a significant problem in any of our simulations.

In the simulations described in the following, networks are trained as auto-associative memories using an artificially constructed alpha-numeric character set represented as a pattern of active units in an 8×8 grid. This set consists of 64 items, eight samples of which are shown in Figure 3. Populations are trained over 500 epochs (presentations of the whole population) using the thermal perceptron algorithm. Attractors are created by auto-associating each item (pattern) with itself as a target output, but on every presentation the input version of the item is distorted by random noise (each unit has a 5% probability of reversing its activation state). In all graphs shown in Figure 3 the results are averaged over 50 replications of the given simulation.

4. Catastrophic Forgetting and Rehearsal

Our illustration of catastrophic forgetting follows the same format as the MLP-based experiments described in Section 2. The network is trained as described previously, on a base population consisting of 44 items (randomly chosen from the character set for each of the 50 replications). After training is complete the network is then trained in the same way (using noise over 500 epochs) on individual items, with 20 of these new items (the remaining characters from the set) introduced in sequence. During this process, the error of the base population is tested after the learning of each new item. The test consists of presenting each base population item (without noise) as an input and letting the network iterate until it reaches a stable state (SS).³ If the input item is itself a SS (no changes occur), then the item is said to be stable. If the network moves from the input item to a different SS, then the item is said to be unstable. In the following figures (Figures 4, 5 and 7), we graph two error measures: (a) the number of base population items which are unstable and (b) the average Hamming distance between the inputs (actual base population items) and the final SSs.

As shown in Figure 4 (no rehearsal condition) the Hopfield-type network is very vulnerable to the catastrophic forgetting effect. After learning even a single new item, the error of the base population increases significantly; after all 20 new items are learned, the ability to recognize the base population has been effectively eliminated.

Rehearsing old base population items as new items are learned alleviates catastrophic forgetting. Figure 4 shows the effect of retraining a randomly selected 25%, 50% or 100% of the base population along with each new item. The more of the base population that gets rehearsed with each new item, the better the retention of the base population; rehearsing the entire base population eliminates forgetting.

5. Pseudorehearsal

While rehearsal provides a simple and direct method of alleviating catastrophic forgetting, as in the MLP case we are interested in the situation where the base population items are not necessarily available for rehearsal. In this section, we show that pseudorehearsal can be successfully adapted to the Hopfield-type network, and that it provides significant protection from catastrophic forgetting. Simulations are based on the learning task described earlier (i.e. a base population of 44 learned items and subsequent learning of a further 20 new items one by one). We briefly describe a range of ways of implementing pseudorehearsal and their effects on catastrophic forgetting.

5.1. Feedforward Pseudoitems

The most direct application of the methods developed in MLP networks is to use pseudoitems generated in the same way (see Section 2). A random input is created (each element has a 50% chance of being on) and fed forward for a single pass (i.e. iterated for a single cycle through the network) to create the associated output. We call these 'feedforward' (FF) pseudoitems.⁴

In our first pseudorehearsal simulations, we used populations of 128 and 256 FF pseudoitems, ruling out duplicates so that each pseudoitem is unique. As each new item is introduced, it is learned not alone, but along with the population of pseudoitems. Pseudoitems are trained without noise. Figure 5 (FF128 and FF256 conditions) illustrates subsequent reduction in error/forgetting (cf. the no rehearsal condition, as for Figure 4). While there is some reduction in both error measures, the error of the base population still climbs steadily as each new item is learned. 256 pseudoitems show little improvement over 128.

5.2. Stable State Pseudoitems

As noted in Section 2, MLP network training can be conceptualized as fitting a function to the base population data points. Pseudoitems sample the function, and relearning the pseudoitems preserves the structure of the function during further learning. Hopfield networks, however, are best interpreted not as function approximators, but as dynamical systems where training manipulates an energy surface to create attractors for stored items. A large number of attractors exist in a trained network, including the actual learned items, and also 'spurious' attractors which do not correspond to learned items but may be generated by interactions between

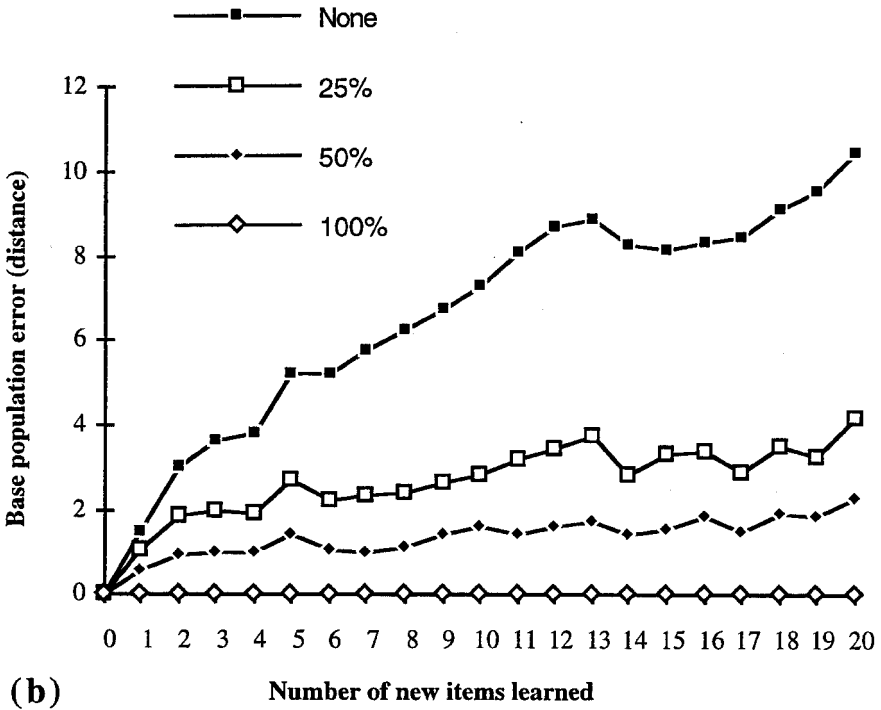
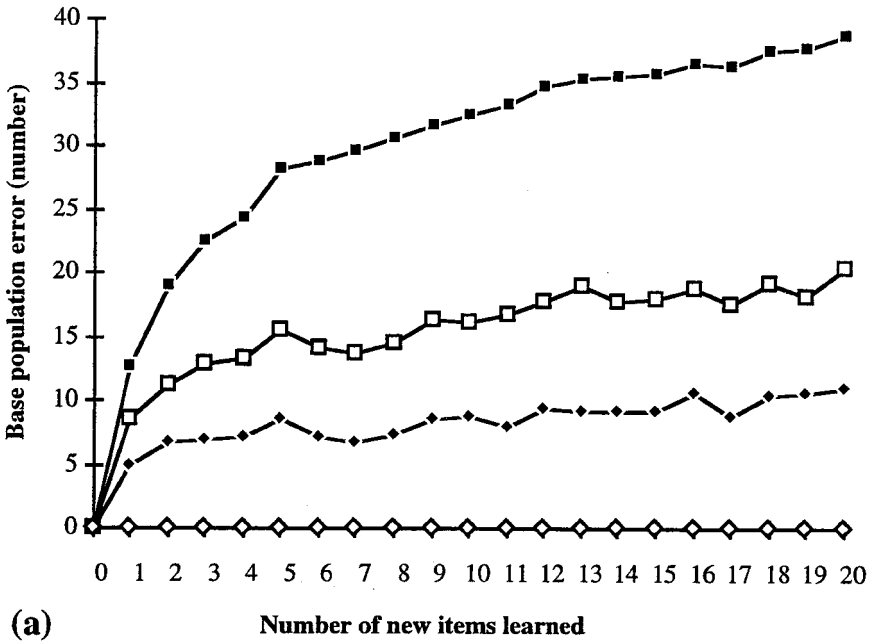


Figure 4. Catastrophic forgetting and rehearsal in a Hopfield-type net. (a) The base population error measure is the number of unstable items. (b) The base population error measure is the average Hamming distance between actual outputs and target base population items.

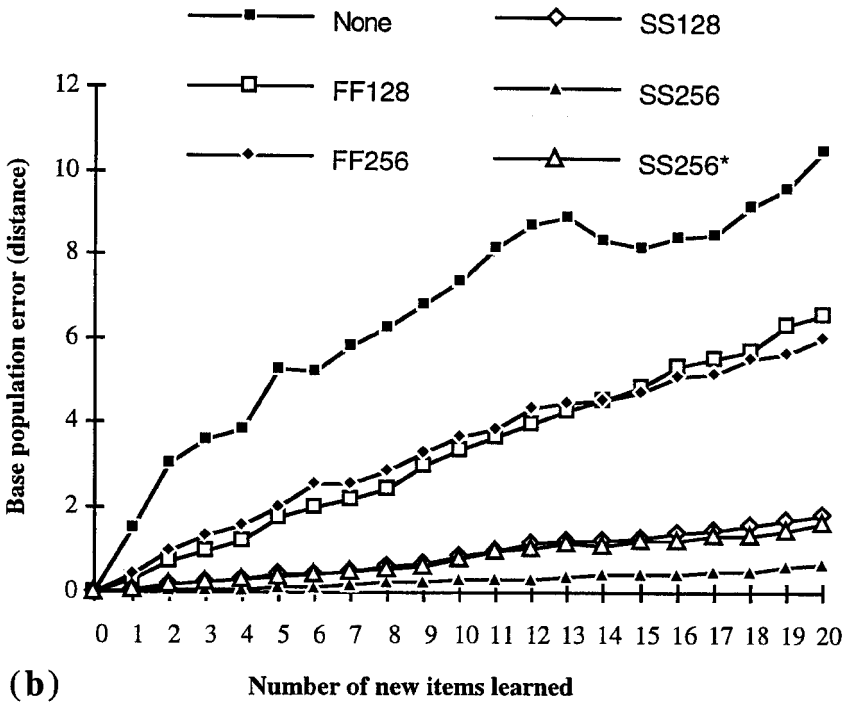
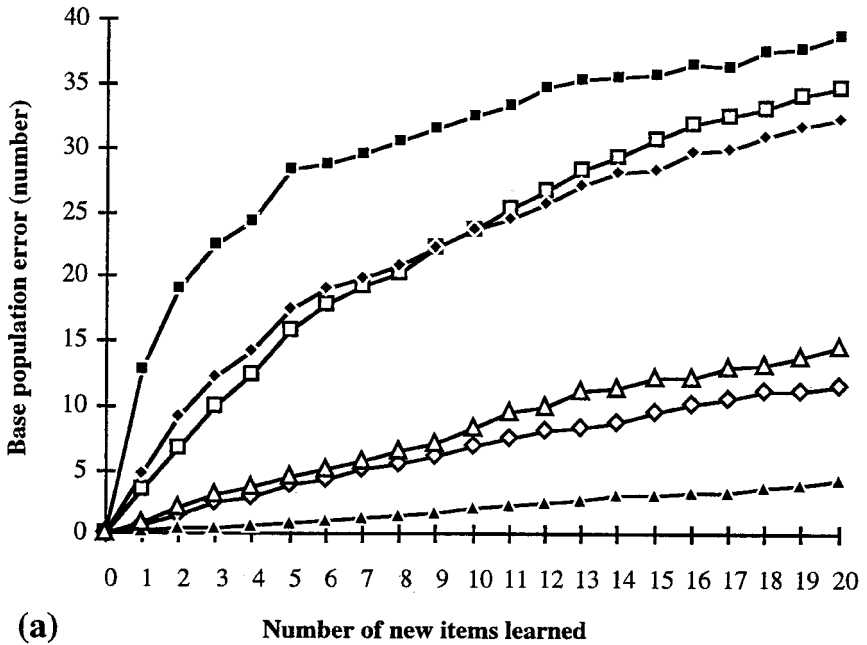


Figure 5. Pseudorehearsal in a Hopfield-type net. (a) The base population error measure is the number of unstable items. (b) The base population error measure is the average Hamming distance between actual outputs and target base population items.

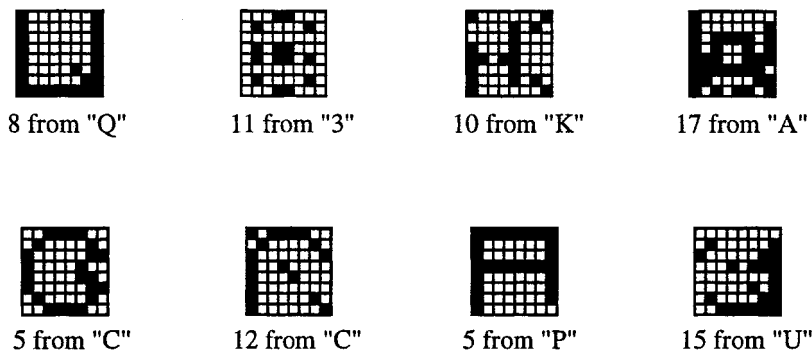


Figure 6. Eight typical SS pseudoitems, showing Hamming distance from nearest base population items.

them. We suggest that locating and preserving attractors, including (or even restricting ourselves to) those which do not correspond to base population items, is an effective way of preserving the attractors that do actually correspond to the previously learned base population.

A new method of creating pseudoitems explores these suggestions. After the base population has been learned, we can create a stable state (SS) pseudoitem by creating a random input and iterating it through the network until a SS/attractor is reached. This final state is used as an auto-associative pseudoitem. A population of unique pseudoitems created in this way can be learned as each new item is introduced, as before. Figure 6 illustrates a selection of typical SS pseudoitems.

Figure 5 illustrates the reduction in error/forgetting using SS pseudoitem populations for three conditions: 128 pseudoitems (SS128), 256 pseudoitems (SS256) and 256 pseudoitems, where items that duplicate actual base population items have been deliberately excluded (SS256*). SS pseudoitems are effective at maintaining good performance on the base population as new times are learned. As in the MLP networks, the number of pseudoitems is significant, the SS256 condition resulting in a significant improvement over the SS128 condition.

Pseudoitem populations generated in this way will naturally contain duplicates of base population items, as these constitute significant attractors in the state space. In our simulations, SS128 populations contained pseudoitems that duplicated roughly 50% of the base population, and SS256 populations duplicated roughly 80% of the base population. If we are assuming no access to the base population for rehearsal, then there is no need, and no principled way, of ruling out these base population duplicates. However, the SS256* condition shows that SS pseudo-rehearsal does not depend on duplicating base population items for its effectiveness.⁵ Although base population duplicates were specifically excluded, these 256 completely novel pseudoitems were roughly as effective as 128 pseudoitems that included a high proportion of base population duplicates. In other words, in the state space created by learning the base population, preserving attractors that do not correspond to base population item attractors (attractors usually described as 'spurious' attractors or 'cross-talk') is effective at preserving the base population item attractors themselves.

We suggest that SS pseudorehearsal is successful because the pseudoitems locate and preserve the attractors created during learning. In the FF case, the only information that a given pseudoitem preserves is the gradient of the energy

surface—the transition from the higher energy state of the pseudo-input to the lower energy state of the pseudo-output—at a single point. Preserving a population of such local gradients has, in our simulations, limited efficacy as a method of preserving the global structure of the energy surface (performance on the base population). Relearning SS pseudoitems, on the other hand, preserves an invariant landmark in the energy surface and also implicitly a whole region of gradient information—the basin of attraction.

5.3. Serial Learning

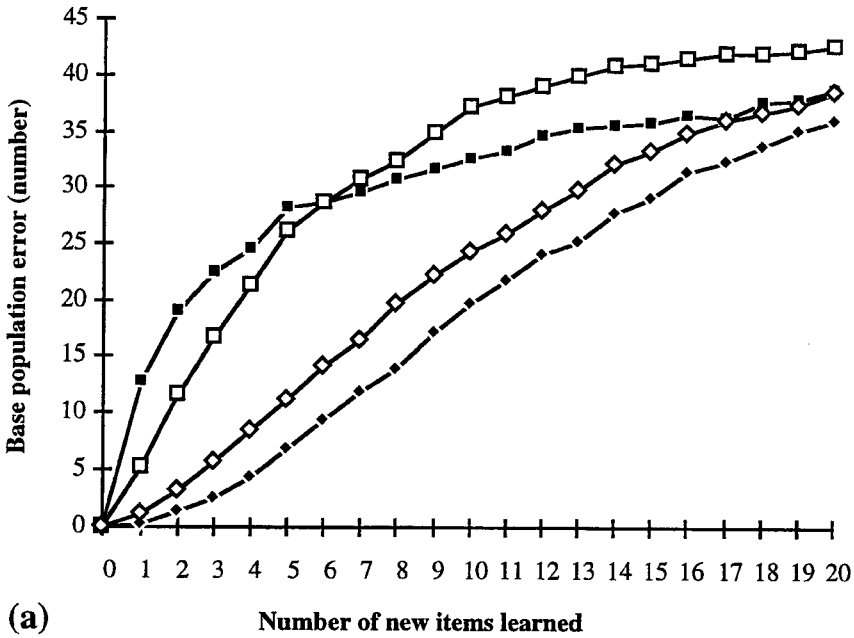
The foregoing simulations use a single pseudoitem population, generated after base population learning is complete, for all subsequent new items. While this provides protection for the base population, it does not prevent new items from interfering with each other. By the time all 20 new items have been learned, retention of the early new items is very poor.⁶ The reason for the vulnerability of the serially learned new items is probably related to a similar effect in MLP networks, namely that single items or small populations are always significantly more vulnerable to catastrophic forgetting than are larger populations which have been learned as a group (Ratcliffe, 1990; Robins, 1995).

In MLP networks, we address this problem successfully by generating a new population of pseudoitems every time a new item is to be learned (this is the method used in the simulations described in Section 2). Each population of pseudoitems protects not only the base population but also previously learned new items. While somewhat effective, this does not work as well in Hopfield-type networks. The interference among new items is indeed reduced significantly, but at the cost of a more rapidly increasing error for the base population. This increased error (for modified training where a new population of pseudoitems is generated prior to learning each new item) is shown in Figure 7 for the FF256, SS256 and SS256* conditions described previously. While SS pseudorehearsal is still significantly better than no rehearsal, the Hopfield-type network is much more sensitive than MLP networks to the compounding of errors permitted by generating successive populations of pseudoitems. The compounding of errors is particularly evident in the FF256 case, where after a certain number of new items, the performance of this method is actually worse than the baseline of no rehearsal. The minor benefit of preserving some local gradient information becomes outweighed by the costs (changes to the energy surface) of the extra learning that is required to do so (effectively three times as many training epochs as the no rehearsal condition).

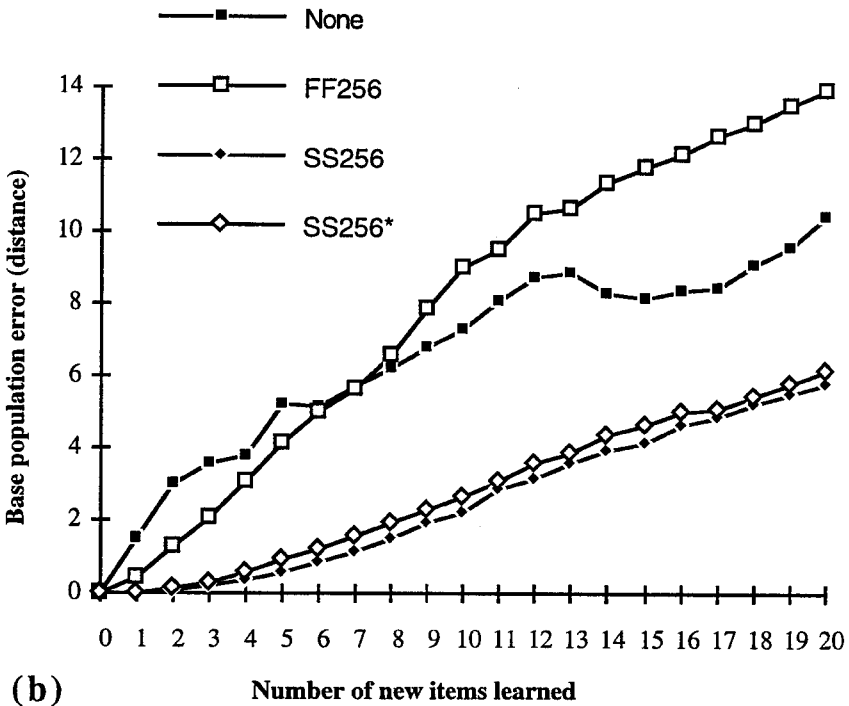
Pseudorehearsal can, however, be highly successful at integrating and preserving the performance of both the base population and new items if the new items are introduced as a single group (rather than serially). In this case, the new items do not sequentially interfere with each other, but converge over a number of iterations on a solution which accommodates all of them (as is the case in the standard concurrent learning of the base population). In our simulations, we created (after base population learning) a single population of SS pseudoitems as usual, and learned it along with all 20 new items as a single group. Good performance was achieved on both the base population (with errors similar to those at the end of serial training in Figure 5) and the new items.

6. Discussion

This initial exploration of pseudorehearsal in Hopfield-type networks has yielded promising practical results. Our exploration of the base population described in



(a)



(b)

Figure 7. Pseudorehearsal in a Hopfield-type net using a new population of pseudoitems for every new item. (a) The base population error measure is the number of unstable items. (b) The base population error measure is the average Hamming distance between actual outputs and target base population items.

this paper has demonstrated significant reductions in catastrophic forgetting as new information is added. We hope to improve this performance in future research.

The main contribution of this paper, however, is to extend the illustration of the principle of pseudorehearsal to a new kind of network. Although the Hopfield-type network and MLPs are very different, the same general mechanisms for maintaining the stability of previously learned information are applicable. Old information can be rehearsed if it is available. If old information is not available, then generating approximations of old information that 'map' the function or behaviour of the network as they are needed, and rehearsing these approximations instead, can also be effective at preserving the actual old information itself.

Two further points emerge from the current study. First, while the principle of pseudorehearsal may be generally applicable, aspects of the specific mechanism employed may need to be adapted for networks of different types. In the Hopfield-type network, SS pseudoitems (that sample the attractors in the state space created by learning) are much more effective than the FF pseudoitems (that are based on the methods used in MLP networks). Second, the pseudoitems generated for pseudorehearsal may naturally include duplicates of actual old information (i.e. actual base population items). There is no need to—and no principled way of—excluding these duplicates. While they are of course beneficial when they occur, the pseudorehearsal mechanism does not depend on these duplicates for its efficacy. The SS pseudoitems used to map the behaviour of the network can consist entirely of the extra attractors created during learning. These attractors are usually described as 'spurious' or 'cross-talk', and regarded as undesirable, interfering with the retention of the trained population items. Our simulations have shown that, in another sense, such attractors can in fact be useful in preserving the learned population.

In future work, we also hope to explore the relationship between the methods outlined in this paper and their opposite, the unlearning of attractors, which several studies have shown also has, in some circumstances, a stabilizing effect on the information stored in Hopfield networks (Christos, 1996; Crick & Mitchison, 1983; Hopfield *et al.*, 1983). There are many open questions regarding these apparently contradictory results, including the effect of the storage capacity of the networks, the role of noise in training and its effect on the kinds of attractors created, the type of learning algorithm and the nature of the learning task.

To summarize in general terms, most ANN learning algorithms are limited to concurrent training methods by excessive plasticity in the representing medium (connection weights) and the consequent problem of catastrophic forgetting. This restriction is undesirable in practical terms, making it difficult to modify or extend any given ANN application without completely retraining the network, and is also highly implausible in terms of cognitive modelling, where so much of human learning is clearly sequential or incremental in nature. Rehearsal and pseudorehearsal are practical methods that successfully address this problem. They enable the on-going or sequential learning of information in ANNs, and consequently also provide a framework for the modelling of lifelong learning/developmental effects in cognition.

Acknowledgements

The authors would like to thank Marcus Frean for discussions on the thermal perceptron algorithm and for comments on drafts of this paper.

Notes

1. The results summarized in Figures 1 and 2 in this section were first presented in Robins (1996). Simulations use the 'Iris' data set (Murphy & Aha, 1994) made up of 150 four-element vectors representing measures taken from individual irises. We used a 4:3:4 auto-associative back-propagation network with a learning constant of 0.05 and a momentum constant of 0.9, an error measure of the average of the sum of the squared errors between output and target vectors, and an error criterion during training of 0.01. All results reported were averaged over 50 individual replications of the simulation (using different populations for each replication). See Robins (1996) for further details.
2. The thermal perceptron rule adds to the standard delta rule an extra term which scales the weight changes so that they tail off exponentially with the absolute value of the total input to the unit. This has the effect of allowing each unit to ignore a large error for any given input, or in other words to 'give up' trying to make a weight change when the required weight change would need to be very large (and therefore likely to disrupt the performance of the unit on all other inputs). This effect is annealed using a temperature parameter, so that the network starts with the standard delta rule and then progressively ignores remaining errors as training progresses. Consequently, the rule finds stable weights for non-separable problems as well as separable ones.
3. A SS/attractor of the network is a state in which updating the activation of every unit does not result in any unit changing its activation state.
4. Note that these pseudoitems are hetero-associative input/output pattern pairs. In contrast, the base population, the new items and the SS pseudoitems described in Section 5.2. are auto-associative patterns, where the input is mapped on to itself.
5. The pseudoitems used in the SS256* condition are on average a Hamming distance of 14.08 from their nearest base population item, so the attractors being used are significantly different from the actual base population items.
6. Even averaged over 50 replications the results are very noisy, and there is no systematic distinction between the pseudorehearsal conditions described above. Early new items have Hamming distance errors in the range of 12–16, and this falls for latter new items to a range of 4–6, with only the final new item being learned accurately.

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